FOROUGH SHIRIN ABKENAR

Ph.D.

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PROFESSIONAL BACKGROUND

Postdoctoral Scholar

University of California Davis

February 2023 – February 2025 Davis, CA

- **Failure Localization:** Developed ML models (FL, DANN, GAN) to detect and classify system failures, improving predictive maintenance; model optimization using knowledge distillation.
- **Multi-Band Upgrade Strategies:** Designed LSTM-based models to optimize network throughput by automating band upgrades.
- **Traffic Engineering:** Explored ML and non-ML methods for maximizing throughput, reliability, and minimizing delay in service provisioning.
- **Mentorship:** Guided Ph.D. students in ML and optical network optimization.

Postdoctoral Scholar

University of California Irvine

February 2022 – February 2023 Irvine, CA

- **Domain Adaptation in ML:** Created ML models for domain adaptation in time-variant datasets, in collaboration with Intel AI/ML.
- **Feature Engineering:** Led feature engineering efforts on complex real-world datasets, including stress classification in the UNITE (health-driven) dataset.
- **Edge Computing:** Developed edge computing frameworks to optimize real-time data processing for IoT applications.

SELECTED PUBLICATIONS

Forough Shirin Abkenar, et al.:

Journal Articles

- "A Survey on Mobility of Edge Computing Networks in IoT: State-of-the-Art, Architectures, and Challenges," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 4, pp. 2329–2365, 2022.
- "ENERGENT: An Energy-Efficient UAV-Assisted Fog-IoT Framework for Disaster Management," *Journal of Communications and Networks*, vol. 24, no. 6, pp. 698–709, 2022.
- "Efficient Task Allocation Protocol for a Hybrid-Hierarchical Spatial-Aerial-Terrestrial Edge-Centric IoT Architecture," *IEICE Transactions*, vol. E105-B, no. 2, pp. 116–130, 2021.

Conference Proceedings

- "Online Domain Adaptive Classification for Mobile-to-Edge Computing," in *IEEE WoWMoM*, Boston, MA, 2023.
- "Selective Data Offloading in Edge Computing for Two-Tier Classification With Local Domain Partitions," in *IEEE PerCom Workshops: Cloud2Things*, Atlanta, GA, 2023.

Full list: GoogleScholar

MY LIFE PHILOSOPHY

"You have to study a great deal to know a little. (Montesquieu)"

EDUCATION

Ph.D. in Electrical and Information Engineering

The University of Sydney

Mar 2018 – Jan 2022

Thesis title: Towards Hyper-efficient IoT Networks Using Fog Paradigm

M.Sc. in IT Engineering

Sahand University of Technology

Sept 2012 – June 2014

B.Sc. in IT Engineering

Shahid Madani University of Azarbayjan

Sept 2008 – June 2012

INTERESTS

Network Optimization, Cloud and Edge Computing, Data Science, AI and Machine Learning, IoT, 5G/6G and Wireless, Optical Communications and Networks, Space Communications (LEO/MEO/GEO satellites)

SKILLS

GAN | CNN | RL | FL | LLM

Python | C++ | SQL | PyTorch | MATLAB | Visual Studio

Problem-Solving | Statistical Analysis | Cross-functional Collaboration

HONORS AND AWARDS

- Generative AI with LLMs Certification, Amazon Web Services, 2025.
- Dean's Award for the Best Thesis, The University of Sydney, 2023.
- CISCO MentorMe Program 2021.